SESSION 6 – Hypothesis Testing (Basic)

Theory Answers (Tasks 1, 3 & 4)

Task 1: Null Hypothesis (H0) and Alternative Hypothesis (H1)

Scenario: You believe that people spend more time on Instagram than on YouTube daily.

Null Hypothesis (H_0): People spend the same amount of time (or less) on Instagram as on YouTube daily. In other words, the average daily time spent on Instagram is less than or equal to the average daily time spent on YouTube.

$$H_0: \mu_{Instagram} \leq \mu_{YouTube}$$

Alternative Hypothesis (H_1): People spend more time on Instagram than on YouTube daily. The average daily time spent on Instagram is greater than the average daily time spent on YouTube.

$$H_1: \mu_{Instagram} > \mu_{YouTube}$$

This is a one-tailed (right-tailed) test because the claim specifically states that Instagram time is greater.

Task 3: Setting up an A/B Test for a Feature

Feature chosen: **New Dark Mode in WhatsApp**

Control and Variant Groups

Control Group (A): Users who continue to see the existing light/default theme of WhatsApp.

Variant Group (B): Users who are shown the new Dark Mode theme.

Metric to Measure

Average daily time spent in the WhatsApp app (or session duration), and optionally the click-through / engagement rate with chats and status features.

Hypotheses

Null Hypothesis (H_0): The new Dark Mode has no effect (or does not increase) on the average daily time spent by users compared to the default theme.

$$H_0: \mu_{Dark\ Mode} \leq \mu_{Default}$$

Alternative Hypothesis (H_1): Users who see the Dark Mode spend more average daily time in the app than users who see the default theme.

$$H_1: \mu_{Dark\ Mode} > \mu_{Default}$$

Task 4: A/B Test Result – New ‘Add to Wishlist’ Button on Flipkart

Out of 1000 users, 520 clicked the new button and 480 clicked the old button (difference of only 40 clicks or 4 percentage points).

This difference is **small** and is quite likely to be due to chance rather than a real improvement. With a sample of 1000 users, a 4% difference is often not large enough to be considered statistically significant at the usual 5% significance level, especially if the underlying click rates are close.

How the p-value helps you decide:

The p-value tells us the probability of observing a difference as large as (or larger than) the one we got, assuming there is actually no real difference between the two buttons (i.e., under the null hypothesis). If the p-value is less than 0.05, we reject the null hypothesis and conclude the new button performs better. If the p-value is greater than 0.05, we do not reject the null hypothesis — the observed difference could easily have occurred by random chance.